

alerts_now.json — Frontend Contract & Field Paths

Root fields:

generated_utc: 2026-02-13T05:24:38.334809+00:00

observer.lat_deg / observer.lon_deg

raw[]: provenance entries {source, path}

counts.total_filtered + counts.by_group{group: count}

groups{group: [item...]}, items[item...] (flat, sorted)

Item (common) fields used by UI:

- Title: item.title
- Group: item.group
- Type: item.type
- Score: item.score_norm (0..1) + item.score_raw
- Time: item.updated_utc (ISO UTC)
- Coords: item.ra_deg, item.dec_deg (deg, nullable)
- Magnitude: item.mag (nullable)
- Notes: item.note (nullable)
- Meta: item.meta (object, group-specific)
- Ingested: item.ingested_utc (ISO UTC)

Wireframe Mapping — Alerts Sidebar (Dock Panel)

Data source: `alerts_now.json.groups[group]` for each tab; fallback: filter `alerts_now.json.items` by group.

Tabs present in contract: keys of `counts.by_group` (render dynamically).

Required tabs in UI v2: GRB, NEOCP, Transient, NEO, Risk, GCN (if present).

Row fields (minimal):

- `icon_12`: derived from `item.group`
- `primary`: `item.title`
- `secondary`: `item.updated_utc` (format UTC->local on client)
- `key metric 1–2`: see below

Key metrics by group (from current payload; optional fields shown in brackets):

- `grb`: `meta.trigger_name` (if exists) / `item.note`
- `neocp`: `item.score_norm` + `item.mag`
- `transient`: `item.type` + (`item.mag` if present)
- `neo`: `meta.dist_ld` (if exists) or `meta.dist_au`; [`meta.diameter_est_km`], [`meta.moid_au`]
- `risk`: `meta.ip` (impact prob), `meta.ps` (Palermo), `meta.last_obs` (if exists)
- `gcn`: `meta.topic` + `meta.kafka_ts_ms` (if exists)

On row click:

- 1) If `item.ra_deg` & `item.dec_deg` not null: center sky-map on coordinates and highlight marker.
- 2) Open Object Detail Modal (or Risk Object Modal if coords missing).
- 3) Keep sidebar selection state (`selected_id`) for keyboard navigation.

Wireframe Mapping — Map Hover Tooltip

Trigger: pointer hover on plotted alert marker.

Lookup: marker stores reference to item.id (or index) in current in-memory alerts list.

Fields:

- icon_32: derived from item.group
 - title: item.title (if group==pha -> red; group==risk -> orange)
 - updated time: item.updated_utc
 - mag: item.mag
 - coords: item.ra_deg / item.dec_deg
 - NEO extra: [meta.diameter_est_km or meta.diameter_km], [meta.moid_au], [meta.dist_id]
- MOID semantics: Minimum Orbit Intersection Distance (Earth MOID), in AU (not a close-approach distance).

Wireframe Mapping — Object Detail Modal (Coords available)

Open from: sidebar row click OR marker click.

Left header: icon_32 (item.group) + item.title.

Top summary chips: score_norm, updated_utc, mag, ra/dec.

Tabs (render only when data present; tabs list is contract-driven):

- 1) Overview: common fields + 1–2 key metrics (same as sidebar).
- 2) Coordinates: RA/DEC, Alt/Az if meta.alt_deg/meta.az_deg exist.
- 3) Source Data: item.source, raw provenance (alerts_now.raw), meta.cad_url_* / meta.topic etc.
- 4) Risk (if group==risk or meta has sentry fields): ip, ps, v_inf, last_obs, impact windows (if present).
- 5) NEO Physical (if meta has sbdb): diameter_km/diameter_est_km, moid_au, H, albedo.
- 6) Payload (GCN): show decoded JSON/text payload safely (collapsed).

Wireframe Mapping — Risk Object Modal (No RA/DEC)

Condition: `item.group=='risk'` AND (`item.ra_deg` is null OR `item.dec_deg` is null).

Behavior: no sky-map centering; instead show 'No coordinates available' panel.

Tabs (always show, but sections may be empty):

- Overview: `item.title`, `updated_utc`, `score_norm`, `note`, `source`.
- Sentry Summary: `meta.*` fields from `alerts_risk.json` (`ip/ps/last_obs` etc.).
- Impacts: if object endpoint provides impact list (future).
- Provenance: raw source path + request params used.

Wireframe Mapping — Alerts Stats Dashboard (New Dialog)

Data source: alerts_now.json.items (current merged view).

Charts:

- 1) Donut: distribution by group = counts.by_group.
- 2) Scatter: NEO close approaches — x=updated_utc (CA time), y=meta.dist_ld (or dist_au) with size by [diameter_est_km], color by pha flag.
- 3) Timeline: alerts volume per hour/day (bucket by updated_utc) for operational awareness.

Controls:

- time window selector (last 24h / 7d / 30d)
- group filter toggles
- export: download current items as JSON